Languages and Machines

state-changes (computer)

① eg: Lang IMC Decision Problems

②

(mathematical objects)

(q0, 0) → (q0)
(q0, 1) → (q1)
(q1, 0) → (q0)
transitions

I/P "state changes"

Decider

"acceptor/rejector"

$$M = (Q, \Sigma, \delta, q0,\ F)$$

$$Q = \{q_0, q_1\}$$

I/P alphabet $\Sigma = \{0, 1\}$

transition function $\delta : Q \times \sum \to Q$

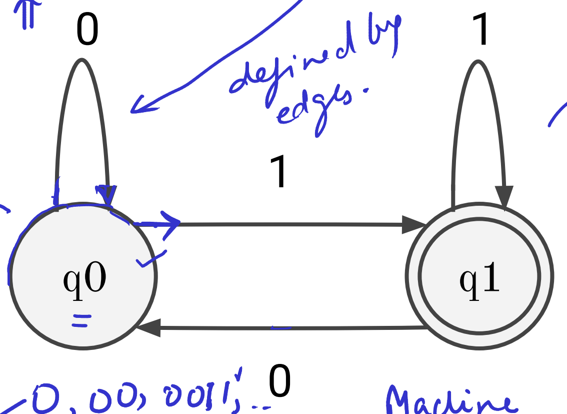
$q0 \in Q$

$F \subseteq Q$

extended T.F $\delta^* : Q \times \sum^* \to Q$

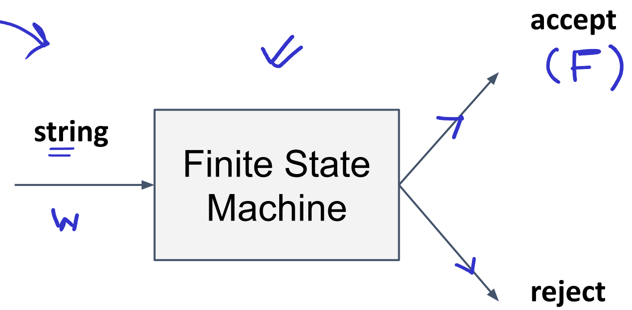
$$L(M) = \left\{ w \in \sum^* : \delta^*(q0, w) \in F \right\}$$

defined by edges.



0, 00, 0011, ..

Machine



Decision Function

Deterministic finite automata

{01, 1, 11, 101, 111, 1001,...}

----1 odd binary integers

"Limited"
↳ No memory

Exercise:

1. Design a FSM for even binary numbers.

{0, 10, 100, 110 ...} $f(x) \to 0, 1$

1. Peter Linz. 1990. An introduction to formal languages and automata. D. C. Heath and Company, USA.

Turing Machine: Model for universal classical computation

$$M = (Q, \Sigma, \Gamma, \delta, q_0, q_{acc}, q_{rej})$$

Q =

Σ = $\{a, b, A, B\}$

Γ $\Delta \in \Gamma$ and $\Sigma \subseteq \Gamma$ (blank)

$$\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$$

q_0, q_{acc}, q_{rej} → $\square$ → q_{acc}, q_{rej}

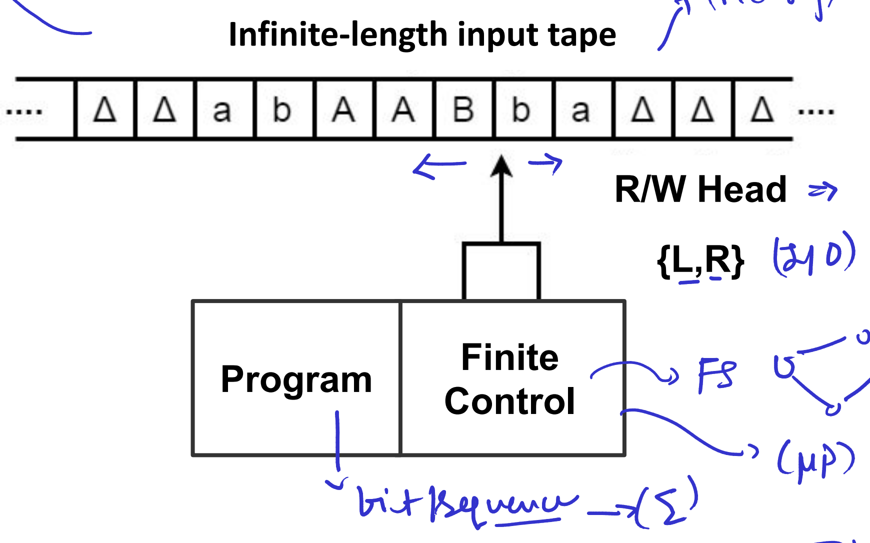
I/P → O/P

TM ⇓ A

UTM

description

TM



FSM

"Programmable Computer"

$x \to \alpha|0\rangle + \beta|1\rangle$

- Effective computation: A TM can simulate any computable function
  - correspondence of algorithm and Turing Machine
- **Universal Turing Machine:** UTM can simulate TMs
- **Church-Turing Thesis:** Every effective method of computation is either equivalent to or weaker than a Turing Machine.

(LC/LC)   (DFA)

"TM can simulate any arbitrary classical computer. Can a TM simulate a quantum computer?"

Exercise:

1. Is there a quantum Turing machine?
2. Is it equivalent to a quantum logic circuit?

1. Peter Linz. 1990. An introduction to formal languages and automata. D. C. Heath and Company, USA.
2. Michael A. Nielsen and Isaac L. Chuang. 2011. Quantum Computation and Quantum Information: 10th Anniversary Edition (10th. ed.). Cambridge University Press, USA.